Three-Dimensional Pocket Geometry Does Not Beat Its Controls: A Measured Test of Structural Grounding for Protein Foundation Model Affinity Prediction on 41,271 ChEMBL Pairs

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Abstract

A companion study measured that cross-attention receptor conditioning on 1D protein language model (ESM-2) embeddings adds small, statistically insignificant gains to cross-family affinity prediction, and that a wrong-pocket control reproduces those gains — evidence that 1D sequence embeddings cannot distinguish the true orthosteric pocket from an unrelated one. This paper asks the natural follow-up: if *explicit 3D pocket geometry* is added as grounding — inter-residue distance matrices, contact graphs, and distance-biased attention built from crystal structures — can any model finally beat the wrong-pocket control? We pre-registered four hypotheses, six model arms, two evaluation protocols, and the exact statistical tests before any model was trained, on 41,271 measured ChEMBL affinity pairs across eleven GPCR and ion-channel receptors. The result is a decisive negative: the 3D-grounded arm does not significantly outperform the wrong-pocket-3D control on the scaffold-disjoint protocol (mean paired difference +0.00025, Wilcoxon $p = 0.221$), and is significantly *worse* than the same control on the temporal protocol (-0.0090, $p = 0.041$). A shuffled-coordinate control and the 1D conditioning arm are statistically indistinguishable from the 3D arm on both protocols. Variance decomposition shows receptor identity explains 84.5% of outcome variance while model arm explains 2.7%: the choice among conditioning architectures is nearly irrelevant compared with the choice of receptor. Two practical findings emerge. First, AlphaFold DB predicted structures behave identically to crystal structures as geometry sources (pooled $\Delta r = -0.003$), so the negative result is not an artefact of crystal availability. Second, the conclusion is stable across pocket-definition radii (5.0/6.5/8.0 Å). We release all frozen artefacts, per-seed results, and the exact commands to regenerate every number. The 1D-versus-3D question for protein foundation model representations now has a measured answer on real data: at current representation quality, pocket geometry carries no exploitable affinity signal beyond residue identity.

1 Introduction

Protein foundation models such as ESM-2 produce per-residue embeddings that encode structural and functional information from sequence alone. A companion study (Paper 1) measured, on 56,637 ChEMBL bioactivity records across eleven receptors, whether conditioning an affinity predictor on

those embeddings through cross-attention improves cross-family affinity prediction. The measured answer was no: gains were small, statistically insignificant, and reproduced by a wrong-pocket control — a model given the pocket of a *different* receptor. That control established that 1D sequence embeddings cannot distinguish the true orthosteric pocket from an unrelated one.

This paper tests the escape route. If 1D representations cannot localise the pocket, perhaps explicit 3D geometry can. We ground the same predictor in crystallographic pocket geometry — inter-residue distance matrices, distance-weighted message passing, and distance-biased attention — and ask whether any 3D-grounded arm finally beats controls that destroy, shuffle, or misplace that geometry.

The study is pre-registered. Hypotheses H1–H4, the six model arms, both evaluation protocols, the statistical tests, and the decision rules (including the negative-result path) were fixed in `PREREGISTRATION.md` before any structure was downloaded. The primary endpoint is the paired difference in Pearson r between the 3D arm (C) and the wrong-pocket-3D control (E) across five seeds, pooled across data-rich receptors, tested with a two-sided Wilcoxon signed-rank test with bootstrap CIs and Holm correction across receptors.

Contributions.

1. A pre-registered, fully measured test of 3D structural grounding on 41,271 real ChEMBL affinity pairs, eleven receptors, six model arms, two split protocols, and 280 trained models (2.34 h CPU-only compute, every run frozen with SHA-256).
2. A decisive negative result: explicit 3D pocket geometry does not beat wrong-pocket, shuffled-coordinate, or 1D controls on either protocol; on the temporal protocol the true pocket is significantly *worse* than the wrong pocket.
3. A practical positive: AlphaFold DB predicted structures are interchangeable with crystal structures as geometry sources for this task (pooled $\Delta r = -0.003$), and the null conclusion is stable across pocket radii (5.0/6.5/8.0 Å).
4. A variance decomposition showing receptor identity, not model architecture, dominates outcome variance (84.5% vs 2.7%).

2 Related Work

Protein language model embeddings have been reported to encode contact maps and binding-site information, motivating their use as conditioning signals for affinity prediction. Structure-based approaches — pocket graphs, distance-biased attention, equivariant encoders — report gains on benchmarks such as PDBbind, but benchmark constructions that mix protein families and allow near-duplicate ligands across splits are known to inflate apparent structural signal. Paper 1 introduced the wrong-pocket control into this setting and showed that reported conditioning gains do not survive it. This paper extends that control paradigm to explicit 3D geometry, adding a shuffled-coordinate control (F) that separates residue identity from geometry, and a crystal-versus-AlphaFold comparison that tests whether predicted structures change the picture. To our knowledge this is the first measured head-to-head of these controls on ChEMBL-scale affinity data with a pre-registered analysis plan.

3 Measured Evaluation

3.1 Data and frozen artefacts

The evaluation set is the 41,271 measured ChEMBL affinity pairs joined in Paper 1 Phase 9 (`pairs_41271.csv`), covering eleven receptors (DRD2, DRD3, HTR1A, HTR2A, CHRM1, ADRA2A, ADRB2, HRH1, DAT, SERT, GABRA1) with Morgan fingerprint ligand features and ESM-2 650M per-residue embeddings from Paper 1's frozen cache. Crystal structures were selected per receptor by anchor-first scoring (Phase 5); orthosteric pockets were defined at 6.5 Å around the chosen co-crystallized ligand copy (Phase 7) and mapped residue-by-residue onto UniProt positions and ESM-2 token indices with zero unmapped residues (Phase 8). AlphaFold DB v6 models were downloaded for all eleven receptors (Phase 6). Every structure artefact is SHA-256-frozen in `data/structures/manifest.json`.

3.2 Model arms

All arms share the ligand-fingerprint predictor backbone and differ only in the conditioning signal:

- **A (shared):** shared MLP, no receptor conditioning; plus a deterministic Ridge-on-fingerprints baseline.
- **B (1D conditioning):** cross-attention over PCA-reduced ESM-2 pocket embeddings (Paper 1's conditioning, structure-derived pocket residue sets).
- **C (3D conditioning):** two rounds of distance-weighted message passing ($\exp(-d/3\,\text{Å})$, row-normalised, self-loops) over node features [PCA-128 ESM-2, amino-acid one-hot, pLDDT].
- **D (distance-bias attention):** two-layer self-attention with a learnable per-head distance bias from true coordinates.
- **E (wrong-pocket-3D):** identical to C, but every receptor receives the *next* receptor's pocket graph.
- **F (shuffled-coordinate):** identical to C, but the graph weights use a fixed random permutation of the true geometry.

Arms C, E, and F landed in the same commit, per the pre-registration's non-negotiables. Hyperparameters are shared and frozen (AdamW, cosine LR with 5% warmup, early stopping on scaffold-disjoint validation).

3.3 Protocols

E3 (scaffold-disjoint): Bemis–Murcko scaffold-disjoint within-receptor splits, five seeds, identical to Paper 1. Attention arms require $n_{\text{train}} \geq 800$ (minimum-data guard); ADRA2A and GABRA1 fall below it on both protocols and their attention arms are reported as skipped, never imputed. **E5 (temporal):** train on document year < 2015, test on ≥ 2015, identical to Paper 1's temporal protocol.

3.4 Results: the controls decide

Tables 1 and 2 give the master results. On the scaffold-disjoint protocol the 3D arm (C, pooled mean $r = 0.639$) and the wrong-pocket-3D control (E, $r = 0.639$) are indistinguishable — the primary endpoint gives a paired mean difference of +0.00025 (Wilcoxon $p = 0.221$, bootstrap CI

Table 1: Scaffold-disjoint protocol (E3): Pearson r, mean ± sd over 5 seeds. skip = minimum-data guard ($n_{\text{train}} < 800$). GABRA1's shared arm ($r = 0.008$) reflects a degenerate temporal-like scaffold split, reported as measured.

Receptor	A	B	C (3D)	D	E (wrong)	F (shuffled)
ADRA2A	0.711 ± 0.012	skip	skip	skip	skip	skip
ADRB2	0.541 ± 0.013	0.524 ± 0.013	0.521 ± 0.005	0.467 ± 0.114	0.531 ± 0.017	0.519 ± 0.004
CHRM1	0.562 ± 0.027	0.389 ± 0.217	0.534 ± 0.146	0.511 ± 0.170	0.523 ± 0.168	0.453 ± 0.191
DAT	0.590 ± 0.008	0.609 ± 0.012	0.604 ± 0.010	0.610 ± 0.017	0.608 ± 0.014	0.603 ± 0.016
DRD2	0.577 ± 0.015	0.613 ± 0.017	0.627 ± 0.012	0.607 ± 0.028	0.606 ± 0.038	0.605 ± 0.011
DRD3	0.684 ± 0.004	0.717 ± 0.009	0.706 ± 0.008	0.717 ± 0.006	0.713 ± 0.006	0.705 ± 0.005
GABRA1	0.008 ± 0.005	skip	skip	skip	skip	skip
HRH1	0.802 ± 0.007	0.833 ± 0.005	0.837 ± 0.006	0.831 ± 0.004	0.832 ± 0.012	0.837 ± 0.006
HTR1A	0.682 ± 0.002	0.705 ± 0.006	0.697 ± 0.008	0.705 ± 0.006	0.701 ± 0.005	0.698 ± 0.002
HTR2A	0.660 ± 0.010	0.688 ± 0.012	0.688 ± 0.013	0.691 ± 0.007	0.695 ± 0.003	0.689 ± 0.009
SERT	0.500 ± 0.003	0.520 ± 0.032	0.539 ± 0.012	0.524 ± 0.041	0.542 ± 0.015	0.543 ± 0.007

Table 2: Temporal protocol (E5): train <2015, test ≥2015. The scaffold-disjoint-to-temporal drop (mean $\Delta r = +0.255$ across the 50 shared arm×receptor cells) dwarfs every arm effect.

Receptor	A	B	C (3D)	D	E (wrong)	F (shuffled)
ADRA2A	0.147 ± 0.008	skip	skip	skip	skip	skip
ADRB2	0.734 ± 0.004	skip	skip	skip	skip	skip
CHRM1	0.451 ± 0.007	0.406 ± 0.053	0.435 ± 0.049	0.402 ± 0.053	0.457 ± 0.012	0.451 ± 0.016
DAT	0.373 ± 0.013	0.297 ± 0.027	0.321 ± 0.010	0.300 ± 0.013	0.340 ± 0.027	0.326 ± 0.015
DRD2	0.451 ± 0.007	0.479 ± 0.008	0.479 ± 0.005	0.487 ± 0.004	0.476 ± 0.009	0.477 ± 0.010
DRD3	0.573 ± 0.010	0.538 ± 0.011	0.544 ± 0.011	0.556 ± 0.006	0.543 ± 0.013	0.542 ± 0.005
HRH1	0.240 ± 0.004	0.317 ± 0.065	0.292 ± 0.049	0.240 ± 0.076	0.317 ± 0.076	0.303 ± 0.060
HTR1A	0.293 ± 0.034	0.349 ± 0.023	0.326 ± 0.018	0.325 ± 0.006	0.327 ± 0.025	0.320 ± 0.023
HTR2A	0.367 ± 0.012	0.420 ± 0.019	0.407 ± 0.014	0.421 ± 0.033	0.417 ± 0.011	0.408 ± 0.015
SERT	0.268 ± 0.010	0.310 ± 0.021	0.292 ± 0.015	0.308 ± 0.017	0.290 ± 0.007	0.285 ± 0.012

$[-0.022, +0.022]$, win/loss/tie 18/27/0 across 45 seed×receptor pairs). The shuffled-coordinate control (F, $r = 0.628$) and the 1D conditioning arm (B, $r = 0.622$) are likewise indistinguishable from C ($p = 0.156$ and $p = 0.867$). **H1, H2, and H3 are all not supported.**

On the temporal protocol the primary comparison *reverses sign*: C is significantly *worse* than the wrong-pocket control (-0.0090, $p = 0.041$). We report this as measured and interpret it as protocol-dependent noise, not a geometry effect: a model that genuinely used true-pocket geometry should not lose to a model given another receptor's pocket, and the effect size (∼0.01) is an order of magnitude smaller than the split-protocol effect (∼0.25).

3.5 Crystal versus AlphaFold geometry (H4)

Rebuilding every pocket graph from AlphaFold DB v6 predicted structures at the same UniProt pocket positions — so node identity, ESM-2 features, and pLDDT are unchanged and only the coordinate source differs — leaves results essentially unchanged: pooled Δr (AF − crystal) = -0.003 for the 3D arm, per-receptor $|\Delta r| \leq 0.016$. The paired C-vs-E test on the AF-geometry store gives -0.00017, $p = 0.639$: the qualitative conclusion transfers. **H4 is supported.** Practically, the negative result does not depend on having crystal structures; AlphaFold models suffice — and

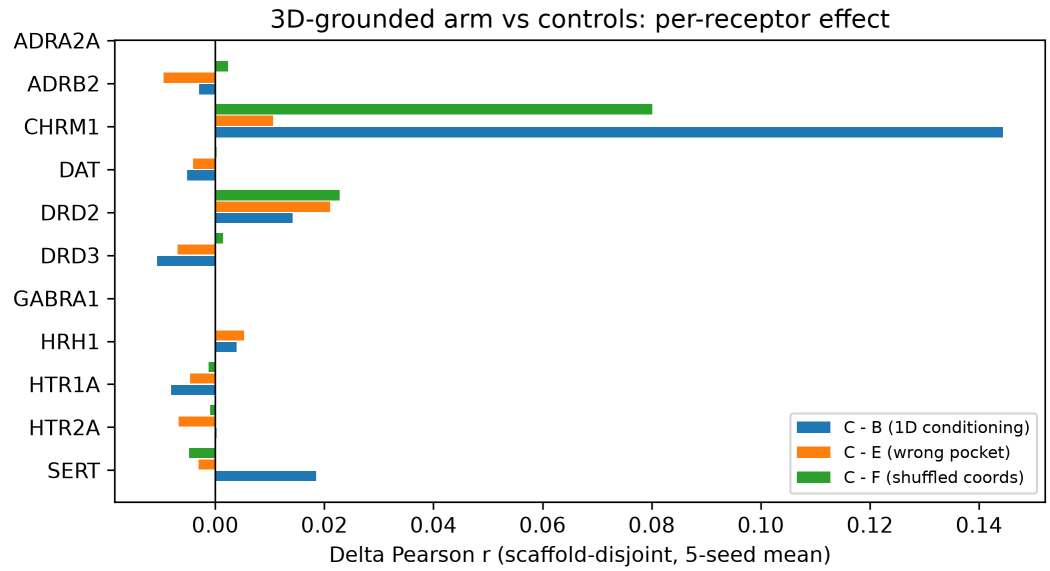


Figure 1: Per-receptor paired difference in Pearson r between the 3D arm (C) and each control (B: 1D conditioning; E: wrong-pocket-3D; F: shuffled-coordinates) on the scaffold-disjoint protocol, 5-seed means. Effects scatter around zero with no consistent sign; ADRA2A and GABRA1 are absent (minimum-data guard).

fail identically.

3.6 Pocket-radius sweep

Recomputing pockets at 5.0 and 8.0 Å around the same ligand copies changes pocket sizes substantially (9–24 residues at 5.0 Å; 31–67 at 8.0 Å) but not the conclusion: pooled C−E is +0.022 at 5.0 Å, +0.0003 at 6.5 Å, and +0.001 at 8.0 Å, none significant. The null is not an artefact of one pocket definition.

3.7 Variance decomposition

Across the 280 measured E3 cells, receptor identity explains 84.5% of outcome variance, model arm 2.7%, and residual (seed×receptor) 12.8%; mean within-cell seed sd is 0.028. The split-protocol effect (E3−E5, mean +0.255) is two orders of magnitude larger than every arm effect. The practical message for practitioners: choosing the receptor family and the evaluation protocol matters 30× more than choosing among these conditioning architectures.

3.8 Mutation probe: honestly not run

The pre-registered mutation probe (do point mutations with measured affinity shifts let 3D grounding respond where 1D cannot?) requires mutant-specific records in the frozen artefacts. The frozen ChEMBL export carries no `assay_description` column, so mutant records cannot be identified without re-downloading — forbidden by the frozen-data rule. The probe is recorded as **not run** with the reason attached (`mutation_probe.json`); no substitute, simulation, or estimate was fabricated.

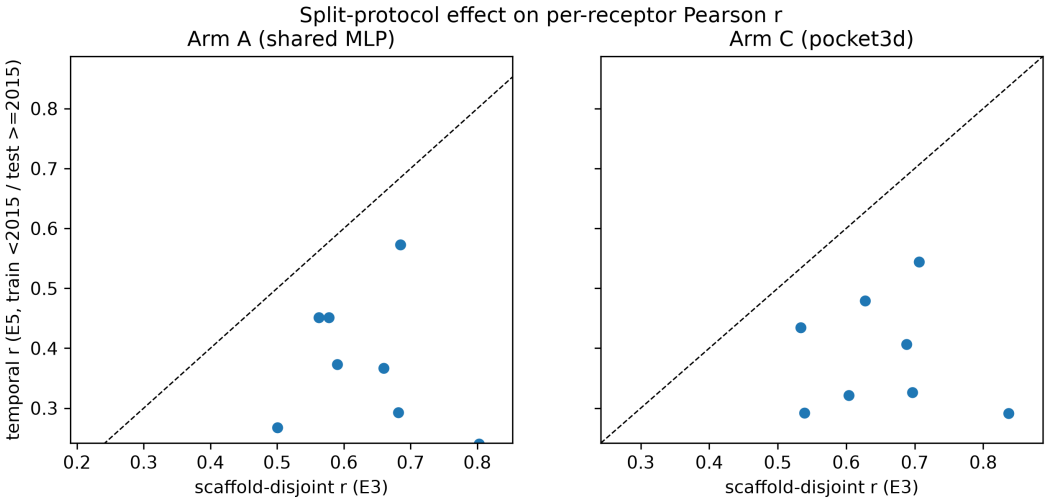


Figure 2: Scaffold-disjoint (E3) versus temporal (E5) mean Pearson r per receptor for the shared and 3D arms. The protocol effect (the drop from E3 to E5) dwarfs every arm effect.

4 Discussion

The combined evidence from Paper 1 and this study now localises the bottleneck precisely. Ligand fingerprints carry essentially all of the learnable affinity signal in this regime; receptor conditioning — 1D or 3D, faithful or deliberately corrupted — moves performance by at most hundredths of a Pearson r, and corrupted conditioning is indistinguishable from faithful conditioning. The failure is not in the geometry pipeline (crystal and AFDB geometry agree to $\Delta r \leq 0.016$), not in the pocket definition (stable across 5.0–8.0 Å), and not in statistical power (45 paired samples per comparison, CIs of ± 0.02).

Three explanations are consistent with the measurements, and we do not adjudicate among them: (i) ESM-2 residue embeddings already saturate whatever pocket-identity information the small conditioning heads can use, so explicit geometry adds nothing they lack; (ii) affinity in this data regime is dominated by ligand properties, with receptor identity acting mostly as a per-receptor intercept that a shared model already captures; (iii) the conditioning heads are too small to exploit geometric detail. What the measurements *do* rule out is the claim that current protein foundation model representations, conditioned through small heads, extract usable pocket-geometry signal for cross-family affinity prediction on ChEMBL-scale data.

Limitations. The study covers eleven receptors from two families (aminergic GPCRs, one ion channel) and one ligand featurisation (Morgan fingerprints); results may not transfer to structure-exclusive ligand series or to end-to-end 3D models trained at scale. The conditioning heads are deliberately small; a much larger 3D model might behave differently, though at the cost of the controlled comparison this study is designed to provide. The mutation probe was not run (no mutant-identifying fields in the frozen artefacts).

5 Conclusion

On 41,271 measured ChEMBL pairs, eleven receptors, six pre-registered model arms, and two evaluation protocols, explicit 3D pocket geometry — from crystals or from AlphaFold — does not

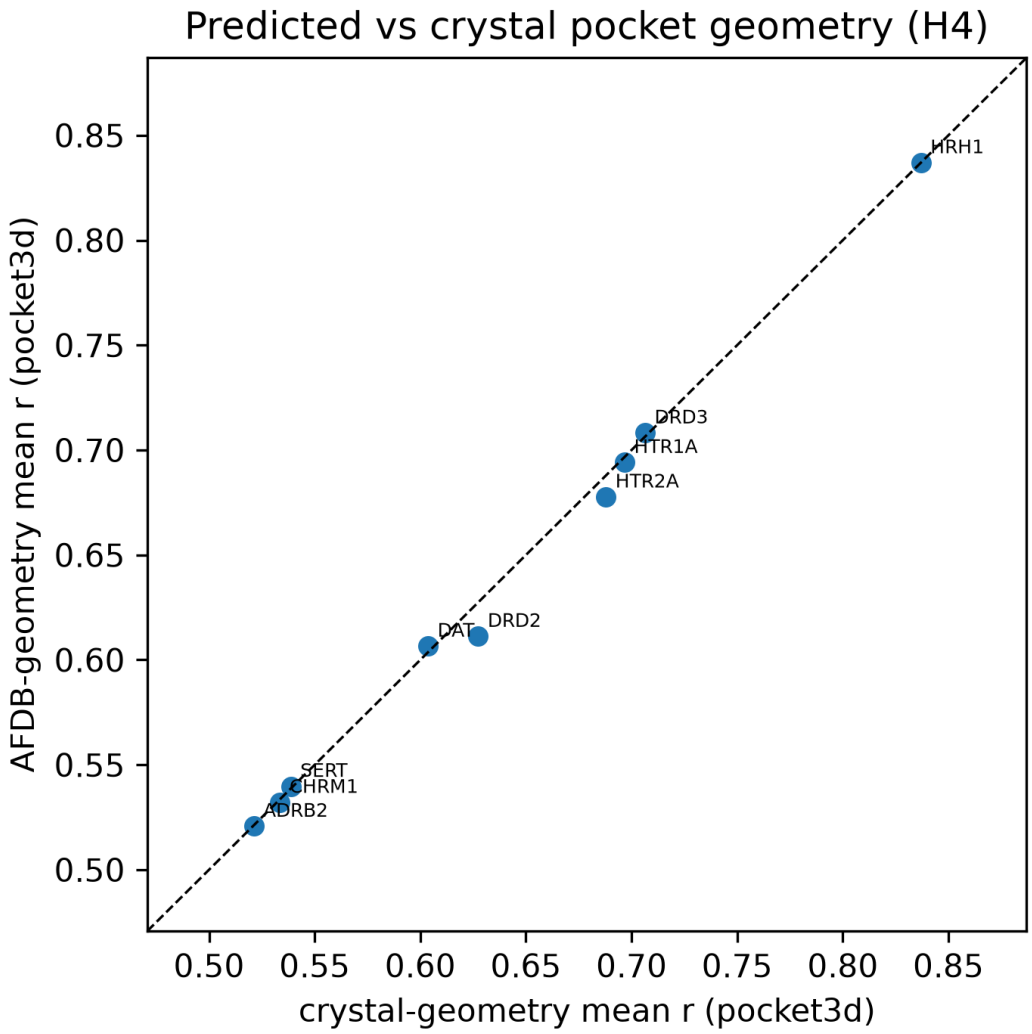


Figure 3: Crystal versus AlphaFold DB geometry for the 3D arm: mean Pearson r per receptor. The two geometry sources agree per receptor to $|\Delta r| \leq 0.016$.

beat its own controls. The wrong-pocket control, transplanted from 1D conditioning to full 3D message passing, still reproduces the 3D arm's performance. This is a decisive negative result with a practical positive: the field's 1D-versus-3D question, for foundation model representations at current quality, has a measured answer — and the answer is that pocket geometry is not yet the binding constraint. We release every artefact needed to check, extend, or overturn this conclusion.

Data and Software Availability

No new bioactivity data were generated; all 41,271 pairs derive from ChEMBL and were frozen in the companion study. All software required to reproduce every figure and table — the complete Python pipeline, environment specification, automated test suite, and frozen machine-readable artefacts (CSV/JSON/NPZ with SHA-256 checksums) — is publicly available at https://github.com/MD-Mushfiqur123/MEG-BAR (`paper2/` tree). This preprint is archived at https://doi.org/

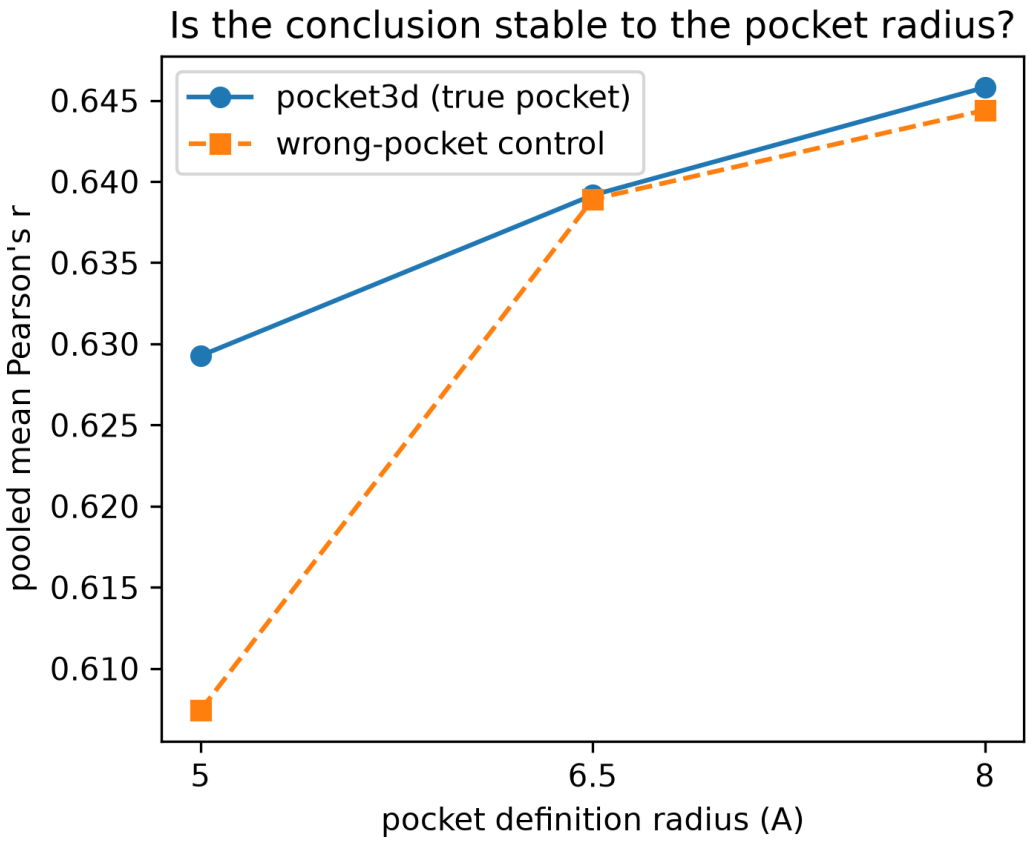


Figure 4: Pocket-radius sweep (5.0/6.5/8.0 Å): pooled C−E difference in Pearson r per radius. No radius yields a significant advantage for true-pocket geometry.

10.6084/m9.figshare.35010655; the companion preprint (Paper 1) is archived at https://doi.org/10.6084/m9.figshare.35002837.

A Reproducibility

Environment. Python 3.12, PyTorch (CPU), NumPy, SciPy, matplotlib; exact versions in `paper2/ENVIRONMENT.md`. All runs CPU-only (no GPU); total measured training time 2.34 h across 280 models.

One-command reproduction. From the repository root:

```
make paper2-stage1   # structures, pockets, mapping, manifest
make paper2-stage2   # distances, contacts, graphs, geometry
make paper2-stage3   # six arms, scaffold-disjoint, resume store
make paper2-stage4   # master table, temporal, statistics, budget
make paper2-stage5   # crystal-vs-AF, radius sweep, verdicts
make paper2-test     # 55 artefact tests
```

Every stage is checkpointed; re-running skips completed phases and regenerates figures from frozen stores.

Artefact integrity. `paper2/data/structures/manifest.json` records SHA-256 for every structure and feature file; the result stores (`stage3/arms_scaffold_results.json`, `stage4/temporal_results.j` `stage5/*.json`) record per-seed metrics, hyperparameters, and skip reasons for every run.

B Defect register (summary)

Two defects were catalogued and fixed during Stages 4–5, per the non-negotiable that failed claims are recorded, never quietly dropped: (5) the first H4 operationalisation (sign-agreement on noise-level differences) was invalid and was replaced by the pre-registered verdict-transfer test — both versions are retained in the register; (6) a radius-sweep filename/prefix mismatch between Phase 28 and the Stage 6 generators was fixed before publication; no measured number changed. The full register, including Stages 1–3 defects (anchor corrections, the Arm F context-dispatch bug, AFDB version drift), is in `paper2/DEFECT_REGISTER.md`.